INFORMATION RETRIEVAL

SEMESTER PROJECT

COURSE CODE: CS444 INSTRUCTOR: DR. ZOYA

TOTAL MARKS: 100

Project Description:

Apply the all given algorithms on each of the following datasets, adapt a preprocessing pipeline to clean these datasets, and compare the results using Accuracy, Precision, Recall and F-Measure metrics and submit the report.

1. Algo – 1: Naive Bayes Classifer
2. Algo – 2: Rocciho Classification
3. Algo – 3: kNN Classification

Datasets for Working:
Following links are for the datasets. The links also contain work details. You have to choose any three datasets to work.

1. https://archive.ics.uci.edu/ml/datasets/Bag+of+Words
2. https://archive.ics.uci.edu/ml/datasets/Health+News+in+Twitter#

Project Report:

The project report should be clear and well-structured, covering all major aspects of the work on the two datasets. Start with a title and abstract summarizing the project, followed by an introduction explaining the problem, objectives, and motivation. Provide a dataset description for both datasets, including their size, classes, and reasons for selection. Include algorithm explanations for Naive Bayes, Rocchio, and kNN with key concepts and formulas. Describe the implementation details, including preprocessing, tools used, parameters, and challenges. Outline the experimental setup, such as train-test split and evaluation methods. Present results using tables with metrics like Accuracy, Precision, Recall, and F1-score for both datasets. Add a discussion and comparison of algorithm performance, followed by a brief conclusion and future work. Finally, include references and an appendix with important code snippets.

Submission Details:

There will be a group comprising three students each. Submit your code on MS Teams by each group leader. The project should have a readme.md file explaining the implementation details as well as detailed steps to run the system. Also submit a project report both in soft form and also in hard form on presentation day. The evaluation of the project will be based on running code and demo in class. The student is required to submit an implementation of the following algorithms in any language of their choice (preferred Python) using machine learning toolkits / API.

Grading:

1. Completion of all modules in Code (50%)
2. Project Report (20 %)
3. Viva (30%)
4. **Note:** marks for the project report and Code may be deducted based on performance in the viva and plagiarism identified.